

Solution for Question 37

Question:

What are the minimum number of 2-to-1 multiplexers required to generate a 2-input AND gate and a 2-input Ex-OR gate?

- (A) 1 and 2
- (B) 1 and 3
- (C) 1 and 1
- (D) 2 and 2

Solution:

A 2-to-1 multiplexer has:

$$Y = \bar{S}I_0 + SI_1$$

where:

- S = Select line
- I_0, I_1 = Inputs
- Y = Output

1. AND Gate using one 2-to-1 MUX

For AND gate:

$$Y = A \cdot B$$

Choose:

$$S = A$$

$$I_0 = 0$$

$$I_1 = B$$

Then:

$$Y = \bar{A}(0) + A(B)$$

$$Y = AB$$

Hence, only **1 multiplexer** is required.

2. EX-OR Gate using 2-to-1 MUX

For XOR gate:

$$Y = A \oplus B$$

Choose:

$$S = A$$

$$I_0 = B$$

$$I_1 = \bar{B}$$

Then:

$$Y = \bar{A}B + A\bar{B}$$

$$Y = A \oplus B$$

To obtain $\bar{B}$, one additional 2-to-1 MUX is required.
Therefore, total multiplexers required:

$$2$$

Final Answer:

(A) 1 and 2
